

Chapter 3 Direct Source QP + MS Screenshots

Cambridge O Level Physics 5054 — Waves and Light up to 3.2.2

Correction: this file uses direct screenshots rendered from the source PDF for BOTH the question-paper pages and the respective mark-scheme answer-key pages. Because the repo PDF contains Paper 1 mark-scheme pages but not Paper 2 mark-scheme pages, this direct-source version uses Paper 1 questions. It still gives 32 relevant exam questions covering the requested syllabus range.

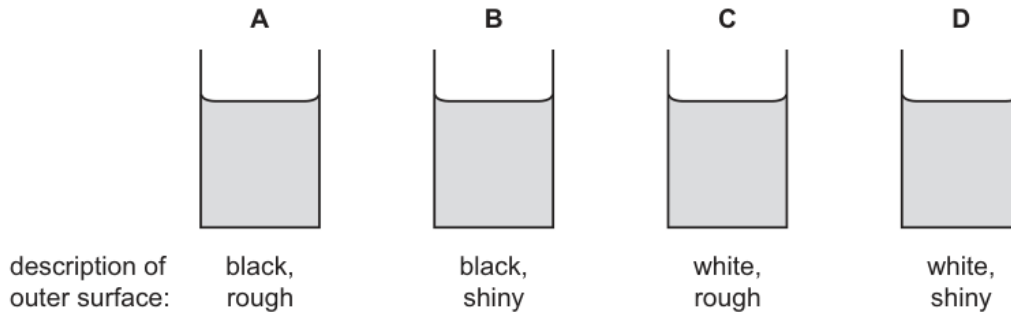
Coverage	Question sources
Wave motion/features/ $v=f\lambda$	A, B, C, D, E, F, G, H
Longitudinal/transverse	A, B, C, F
Reflection/refraction/diffraction/TIR/optical fibre	A, B, C, D, E, F, G, H
Plane mirror/reflection of light	A, C, D, F, G

A: O/N 2025 Paper 1 Variant 1 — sound waves, plane mirror, refraction, optical fibre — QP screenshot

- 17** Four metal cans are identical except for the colour and the texture of their outer surfaces.

100 cm³ of water at 70 °C is poured into each can.

Which can cools the most rapidly?



- 18** The sentences describe sound waves. Three words have been omitted.

Sound waves are longitudinal waves that consist of regions of higher pressure called1..... and regions of lower pressure called2..... .

An echo occurs when a sound wave is3..... .

Which words complete gaps 1, 2 and 3?

	1	2	3
A	compressions	rarefactions	reflected
B	compressions	rarefactions	refracted
C	rarefactions	compressions	refracted
D	rarefactions	compressions	reflected

- 19** A ray of light strikes a plane mirror at an angle of incidence of 20°.

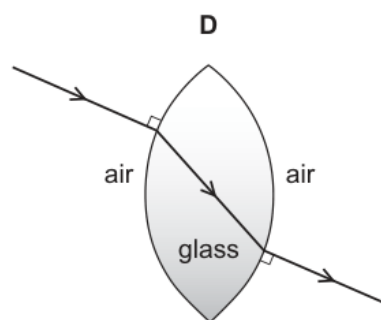
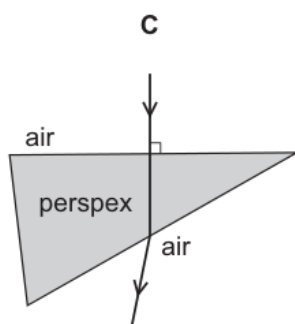
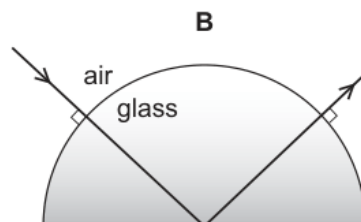
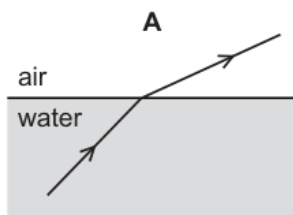
The angle of incidence is then increased by 5°.

What is the new angle between the incident ray and the reflected ray?

- A** 10° **B** 25° **C** 45° **D** 50°

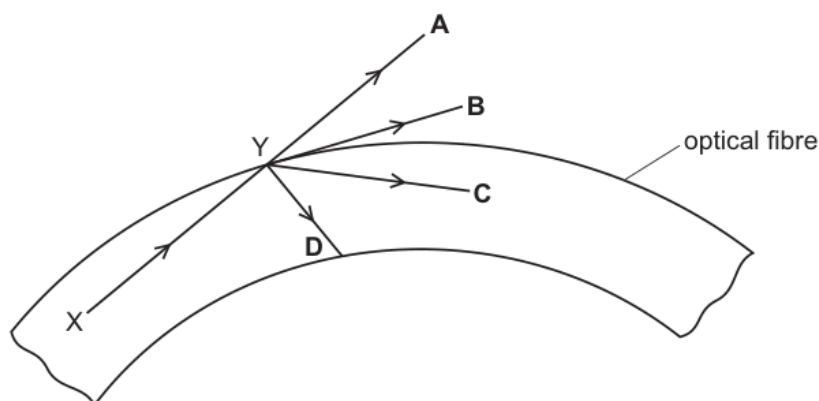
A: O/N 2025 Paper 1 Variant 1 — sound waves, plane mirror, refraction, optical fibre — QP screenshot

22 In which diagram is the path of the light ray **not** correct?



23 A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

In which direction does the ray of light travel after reaching point Y?



24 A firework is launched vertically from the ground.

When the firework reaches a height of 360 m, the firework explodes, producing coloured light.

A student on the ground below the firework incorrectly measures the time between seeing the coloured light and hearing the explosion. The student obtains a value of 1.8 s.

What is the speed of sound that the student's value suggests?

- A** 200 m/s **B** 360 m/s **C** 400 m/s **D** 650 m/s

A: O/N 2025 Paper 1 Variant 1 — sound waves, plane mirror, refraction, optical fibre — QP screenshot

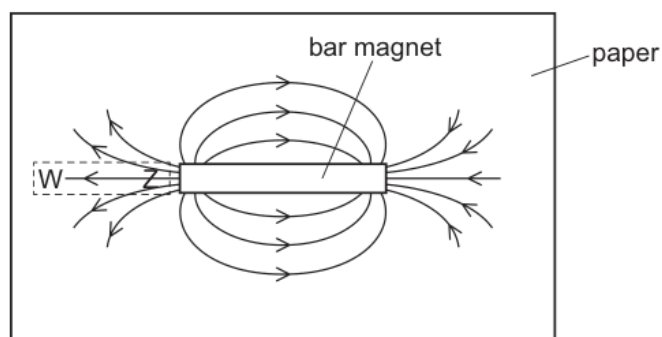
- 25** A loudspeaker produces a sound wave of frequency 50 Hz.

The amplitude of the sound wave is increased.

What is heard?

- A** a louder sound of a higher pitch
 - B** a louder sound of the same pitch
 - C** a sound of higher pitch but the same loudness
 - D** a sound of the same pitch and same loudness
- 26** A bar magnet is placed on a sheet of paper.

The diagram shows the field lines due to the magnet.



A soft-iron bar is placed on the paper on the rectangle marked WZ. The soft-iron bar becomes an induced magnet.

What are the magnetic poles at the ends of the soft-iron bar?

	pole at W	pole at Z
A	N	N
B	N	S
C	S	N
D	S	S

- 27** Which material is an electrical conductor at room temperature?

- A** aluminium
- B** glass
- C** plastic
- D** rubber

A: O/N 2025 Paper 1 Variant 1 — sound waves, plane mirror, refraction, optical fibre — respective MS screenshot

Question	Answer	Marks
1	C	1
2	D	1
3	C	1
4	C	1
5	B	1
6	B	1
7	A	1
8	C	1
9	C	1
10	A	1
11	B	1
12	D	1
13	C	1
14	D	1
15	A	1
16	B	1
17	A	1
18	A	1
19	D	1
20	D	1
21	A	1
22	D	1
23	C	1
24	A	1
25	B	1
26	B	1
27	A	1
28	A	1

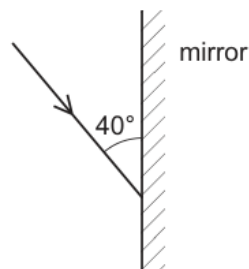
B: M/J 2025 Paper 1 Variant 2 — ripple tank refraction, reflection angles, critical angle, ultrasound — QP screenshot

- 20** A water wave travels from deep water to shallow water in a ripple tank.

What happens to the speed and to the wavelength of the wave?

	speed of wave	wavelength
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

- 21** The diagram shows a ray of light directed at a plane mirror.

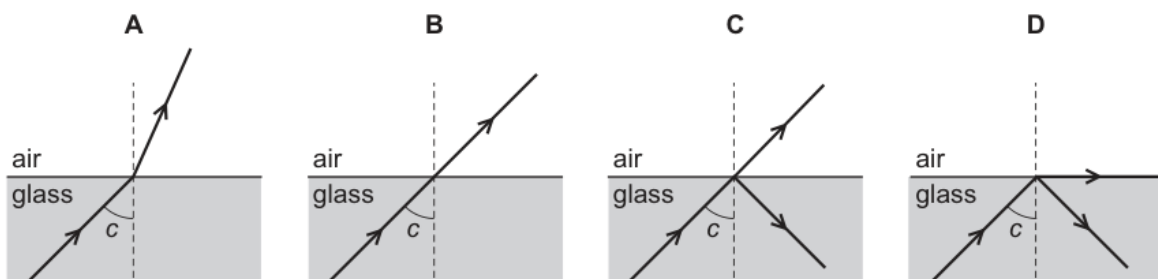


What are the angle of incidence and the angle of reflection?

	angle of incidence / °	angle of reflection / °
A	40	40
B	40	50
C	50	40
D	50	50

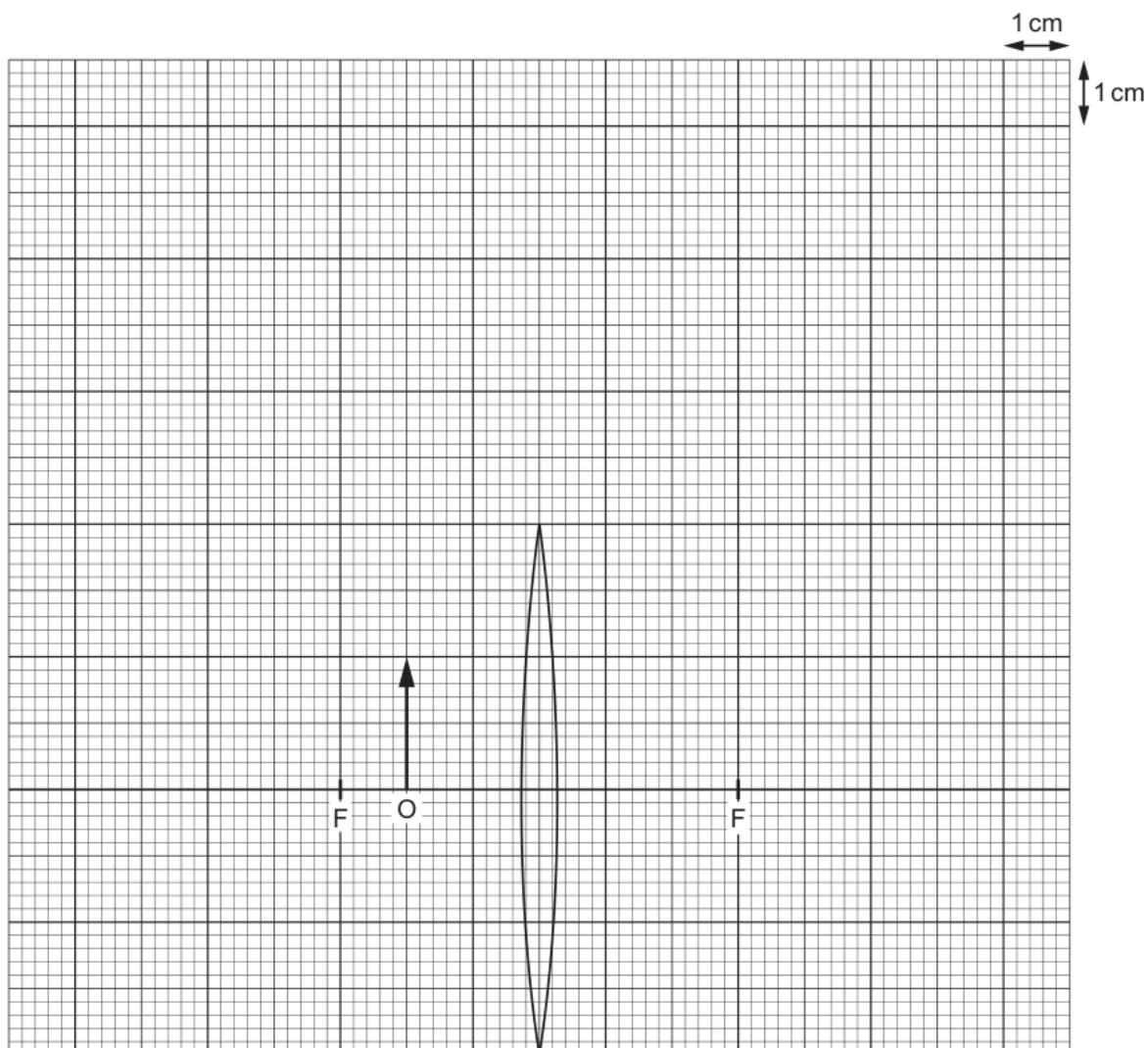
- 22** A ray of light in glass is incident on the surface at an angle c . The angle c is the critical angle.

Which diagram shows what happens to the light?



B: M/J 2025 Paper 1 Variant 2 — ripple tank refraction, reflection angles, critical angle, ultrasound — QP screenshot

- 23 An object O is placed between a converging lens and one principal focus F of the lens.



What is the magnification of the image formed?

- A** 1.0 **B** 2.0 **C** 3.0 **D** 6.0
- 24 Which statement about red light and violet light is correct?
- A** A prism deviates red light more than it deviates violet light.
- B** Red light has a lower frequency than violet light.
- C** Red light has a shorter wavelength than violet light.
- D** The speed of red light in a vacuum is less than the speed of violet light in a vacuum.

B: M/J 2025 Paper 1 Variant 2 — ripple tank refraction, reflection angles, critical angle, ultrasound — QP screenshot

- 25** Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

- A** optical fibre communication
- B** prenatal scanning
- C** photography
- D** telephone communications

- 26** A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m/s.

What is the distance of the whale above the sea bed?

- A** 300 m **B** 450 m **C** 600 m **D** 750 m

- 27** Which row describes ultrasound waves?

	type of wave	frequency
A	longitudinal	smaller than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	smaller than 20 kHz
D	transverse	greater than 20 kHz

- 28** A quantity is defined as the work done by a source in moving unit charge around a complete circuit.

What is the quantity and what is its unit?

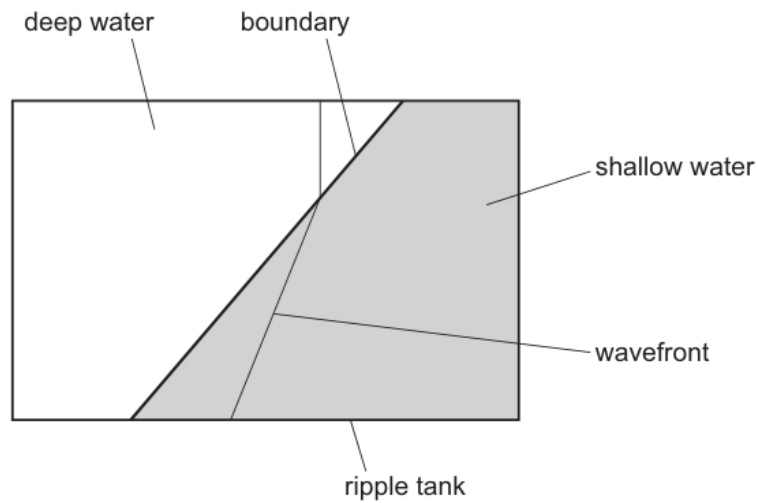
	quantity	unit
A	electromotive force	N
B	electromotive force	V
C	potential difference	N
D	potential difference	V

B: M/J 2025 Paper 1 Variant 2 — ripple tank refraction, reflection angles, critical angle, ultrasound — respective MS screenshot

Question	Answer	Marks
1	B	1
2	D	1
3	D	1
4	B	1
5	A	1
6	D	1
7	D	1
8	C	1
9	A	1
10	C	1
11	A	1
12	D	1
13	D	1
14	B	1
15	D	1
16	A	1
17	B	1
18	C	1
19	C	1
20	D	1
21	D	1
22	D	1
23	C	1
24	B	1
25	B	1
26	A	1
27	B	1
28	B	1

C: O/N 2021 Paper 1 Variant 2 — water-wave refraction, transverse refraction, plane mirror, echoes — QP screenshot

- 21** The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.

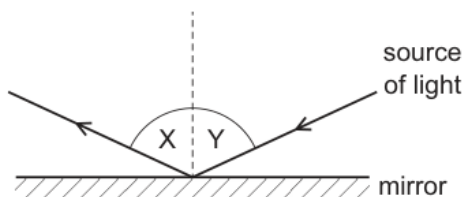


What happens as the wavefront moves into the shallow water?

- A** The speed of the wavefront increases.
 - B** The speed of the wavefront decreases.
 - C** The wavelength of the wave remains constant.
 - D** The wavelength of the wave increases.
- 22** Which example describes a transverse wave being refracted?
- A** infrared waves travelling in an optical fibre
 - B** light waves from an object entering a magnifying glass
 - C** ultrasound waves being used to clean jewellery
 - D** water waves hitting a barrier

C: O/N 2021 Paper 1 Variant 2 — water-wave refraction, transverse refraction, plane mirror, echoes — QP screenshot

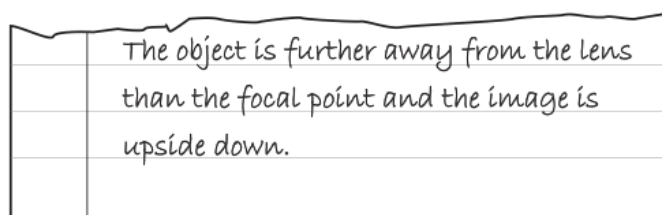
- 23 Light reflects from a plane mirror as shown.



Which row is always correct?

	symbol of angle X	relationship between X and Y
A	i	$X = Y$
B	i	$X + Y = 90^\circ$
C	r	$X = Y$
D	r	$X + Y = 90^\circ$

- 24 A piece of paper torn from an exercise book is shown.

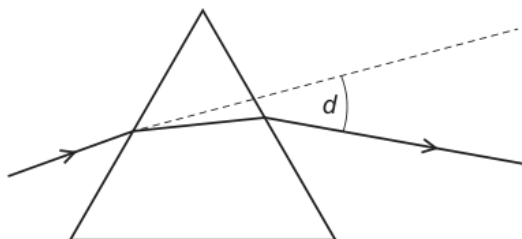


Which process is being described?

- A** the formation of a virtual image by a diverging lens
B the formation of a virtual image by a converging lens
C the formation of a real image by a diverging lens
D the formation of a real image by a converging lens
- 25 A girl is long-sighted.
- Which statement is correct?
- A** She sees close objects less clearly than a person with normal vision.
B She sees distant objects more clearly than a person with normal vision.
C The fault is corrected with a diverging lens.
D The image of a close object is formed in front of her retina.

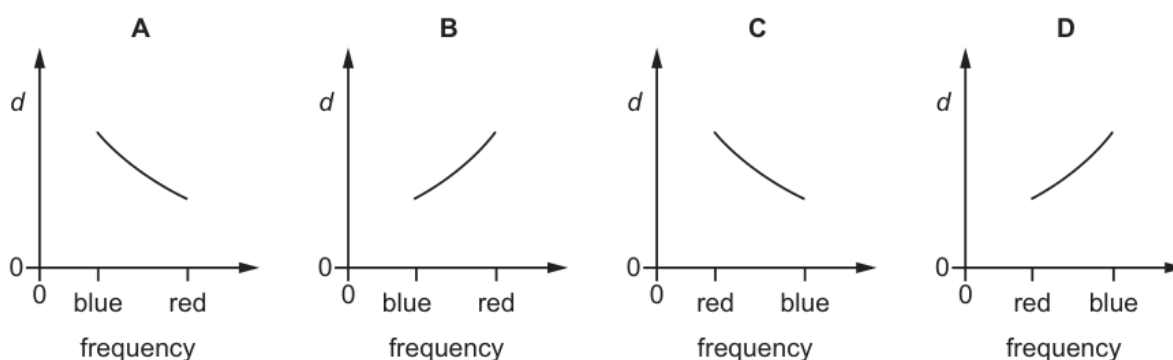
C: O/N 2021 Paper 1 Variant 2 — water-wave refraction, transverse refraction, plane mirror, echoes — QP screenshot

- 26 Light rays are deviated by a prism.



The deviation angle d is measured for light rays of different frequency, including blue light and red light.

Which graph of d against frequency is correct?



- 27 A satellite orbits the Earth at a height of 300 km above the surface. The speed of electromagnetic waves is 3.0×10^8 m/s.

What is the minimum time taken for a radio signal transmitted from the Earth to reach the satellite?

- A** 1.0×10^{-6} s **B** 2.0×10^{-6} s **C** 1.0×10^{-3} s **D** 2.0×10^{-3} s

- 28 Which of the following is most closely associated with echoes?

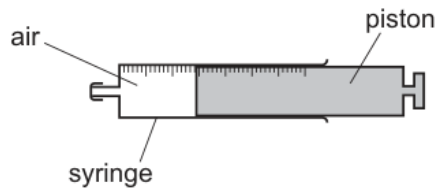
- A** focussing
B reflection
C refraction
D total internal reflection

C: O/N 2021 Paper 1 Variant 2 — water-wave refraction, transverse refraction, plane mirror, echoes — respective MS screenshot

Question	Answer	Marks
1	C	1
2	B	1
3	B	1
4	D	1
5	D	1
6	C	1
7	A	1
8	D	1
9	B	1
10	D	1
11	B	1
12	C	1
13	A	1
14	D	1
15	B	1
16	B	1
17	B	1
18	D	1
19	B	1
20	D	1
21	B	1
22	B	1
23	C	1
24	D	1
25	A	1
26	D	1
27	C	1
28	B	1

D: O/N 2019 Paper 1 Variant 2 — amplitude, wavefronts, plane mirror, refraction angle — QP screenshot

- 20** A syringe containing air is sealed at one end. The piston is free to move.



The air is heated.

What happens to the average distance between air molecules and the density of the air?

	average distance between air molecules	density of the air
A	increases	decreases
B	increases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

- 21** A metal disc is heated to 600°C and then lowered into a beaker of water.

What happens to the mass of the metal disc and to the mass of the water in the beaker?

	mass of metal disc	mass of water in beaker
A	increases	decreases
B	increases	no change
C	no change	decreases
D	no change	increases

- 22** Which description defines the amplitude of a water wave in a ripple tank?

- A** half of the vertical distance between a trough and a peak
- B** the horizontal distance between adjacent peaks
- C** the horizontal distance between a trough and an adjacent peak
- D** the vertical distance between a trough and a peak

D: O/N 2019 Paper 1 Variant 2 — amplitude, wavefronts, plane mirror, refraction angle — QP screenshot

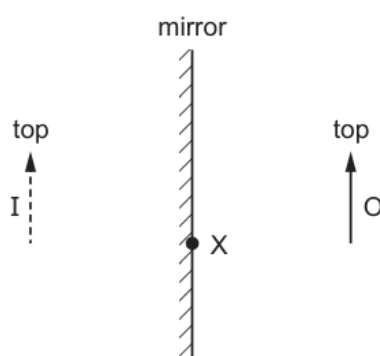
23 Here are four statements about wavefronts.

- 1 Wavefronts from a point source are straight lines.
- 2 Wavefronts from a point source are circular.
- 3 Wavefronts are parallel to the direction of energy movement.
- 4 Wavefronts are perpendicular to the direction of energy movement.

Which two statements are correct?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

24 An object O is placed in front of a plane mirror. I is the image formed.



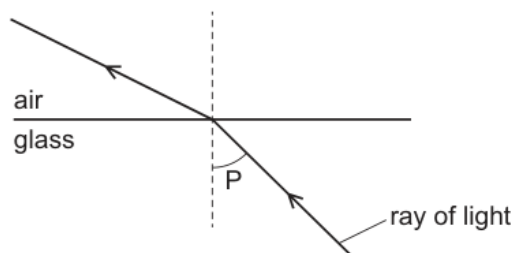
A ray from the top of the object is incident on the mirror at X.

What happens to this ray?

- A** It reflects and passes through the bottom of O.
B It reflects and passes through the top of O.
C It reflects as though it came from the bottom of I.
D It reflects as though it came from the top of I.

D: O/N 2019 Paper 1 Variant 2 — amplitude, wavefronts, plane mirror, refraction angle — QP screenshot

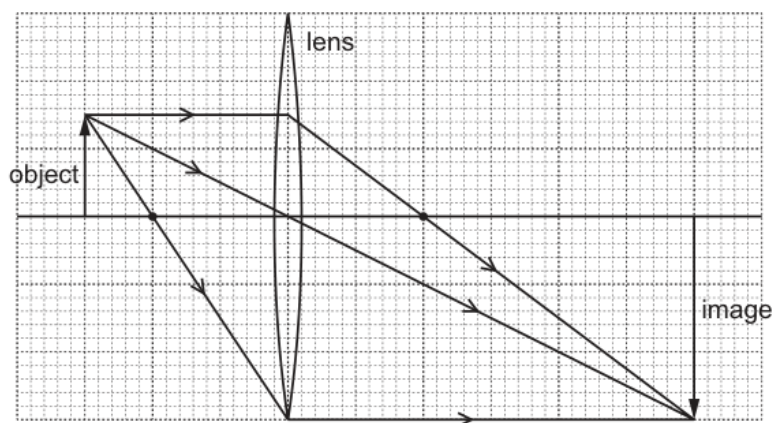
- 25 The diagram shows light passing from glass into air.



What is the name of angle P?

- A the angle of incidence
 - B the angle of reflection
 - C the angle of refraction
 - D the critical angle
- 26 An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

The arrangement is shown on the scale diagram.



What is the linear magnification produced by the lens?

- A 0.50
- B 1.5
- C 2.0
- D 6.0

D: O/N 2019 Paper 1 Variant 2 — amplitude, wavefronts, plane mirror, refraction angle — respective MS screenshot

Question	Answer	Marks
1	B	1
2	C	1
3	B	1
4	D	1
5	C	1
6	D	1
7	B	1
8	C	1
9	D	1
10	D	1
11	D	1
12	A	1
13	C	1
14	D	1
15	D	1
16	C	1
17	B	1
18	C	1
19	A	1
20	A	1
21	C	1
22	A	1
23	D	1
24	D	1
25	A	1
26	C	1
27	C	1
28	D	1

E: O/N 2018 Paper 1 Variant 2 — wave definitions, dispersion, total internal reflection — QP screenshot

- 21 Two liquid-in-glass thermometers are almost identical. They contain the same quantity of the same liquid and the diameters of their capillary tubes are the same.

The only difference is that one thermometer is longer than the other.

Which row shows how the ranges and the sensitivities of the thermometers compare?

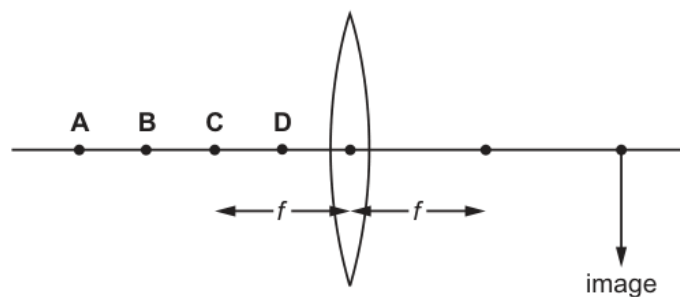
	ranges	sensitivities
A	different	different
B	different	same
C	same	different
D	same	same

- 22 Which statement about a water wave is correct?

- A** The amplitude is the vertical distance between a trough and a peak.
- B** The frequency is the number of troughs passing a point in one second added to the number of peaks passing a point in one second.
- C** The speed is the horizontal distance travelled per second by a peak.
- D** The wavelength is the horizontal distance between a trough and a peak.

- 23 The diagram shows a thin converging lens of focal length f .

Where must an object be placed to produce a real image in the position shown?



- 24 White light is dispersed by a prism. Compared with blue light, the **red** light is

- A** slowed down less and refracted less.
- B** slowed down less and refracted more.
- C** slowed down more and refracted less.
- D** slowed down more and refracted more.

E: O/N 2018 Paper 1 Variant 2 — wave definitions, dispersion, total internal reflection — QP screenshot

- 25 A student reads the following in her physics book.

'The incident angle is greater than 42° which is the critical angle for glass in air.'

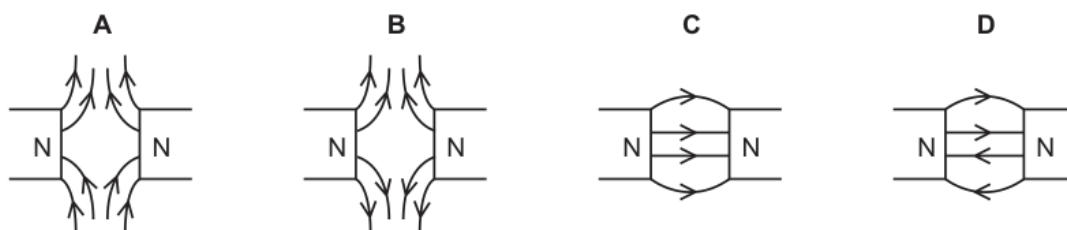
What is the student reading about?

- A focal length of a glass lens
 - B reflection in a plane mirror
 - C refraction as light enters glass
 - D total internal reflection
- 26 Which statement about radio waves is correct?
- A Radio waves are sound waves.
 - B Radio waves are used to kill cancerous cells.
 - C Radio waves are used in television communications.
 - D Radio waves have frequencies higher than those of visible light.
- 27 Ultrasound is used to map the ocean floor.

During one survey, the depth of water is 1200 m. An ultrasound pulse is sent from the surface and when it returns to the ship, another pulse is sent immediately. In any period of 8.0 s, five pulses are sent down from the surface.

What is the speed of the ultrasound in water?

- A 150 m/s
 - B 300 m/s
 - C 750 m/s
 - D 1500 m/s
- 28 Which diagram shows the magnetic field pattern in the region between the N-poles of the two bar magnets?



E: O/N 2018 Paper 1 Variant 2 — wave definitions, dispersion, total internal reflection — respective MS screenshot

Question	Answer	Marks
1	C	1
2	A	1
3	B	1
4	B	1
5	A	1
6	D	1
7	C	1
8	D	1
9	B	1
10	A	1
11	D	1
12	C	1
13	D	1
14	B	1
15	B	1
16	C	1
17	A	1
18	D	1
19	C	1
20	C	1
21	B	1
22	C	1
23	A	1
24	A	1
25	D	1
26	C	1
27	D	1
28	B	1

**F: O/N 2017 Paper 1 Variant 2 — longitudinal wave, reflection, Snell's law, TIR
— QP screenshot**

- 17 A thermocouple is a type of thermometer.

Which statement applies to a thermocouple?

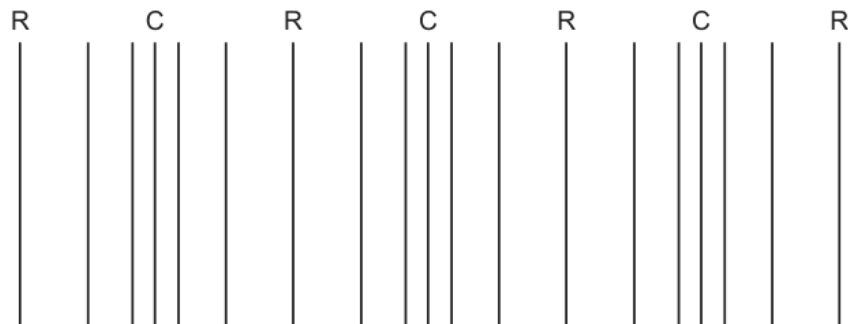
- A It can measure rapidly varying temperatures.
- B It must be made from three different metals.
- C Its resistance decreases when one junction is heated.
- D The distance between the two junctions increases as the temperature increases.

- 18 A heater is designed to radiate thermal energy.

Which change to the design decreases the thermal energy emitted by radiation?

- A a darker coloured surface
- B a higher surface temperature
- C a larger surface area
- D a shinier surface

- 19 A sound wave in a solid is represented by a series of lines. The diagram shows compressions C and rarefactions R at one instant in time.

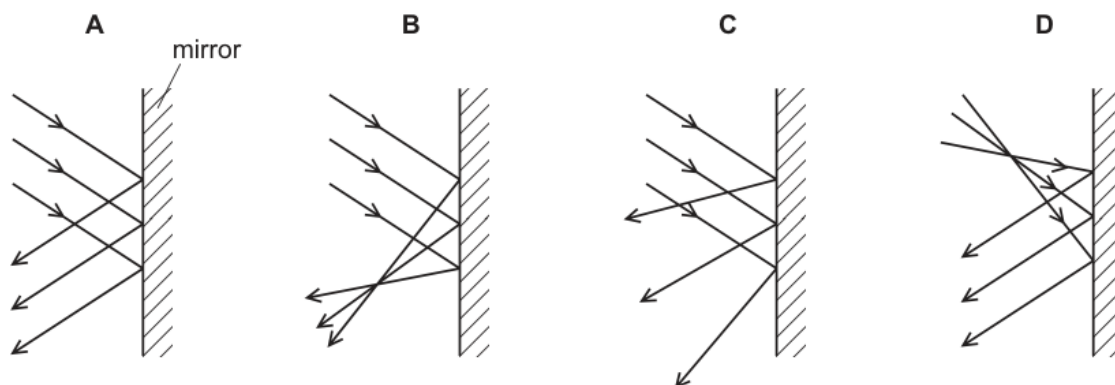


Which statement is correct?

- A All particles on one line move in the same direction at the same time.
- B The direction of travel of the sound wave is parallel to the lines.
- C The distance between a compression and its adjacent rarefaction is a wavelength.
- D The pattern of lines represents a transverse wave.

F: O/N 2017 Paper 1 Variant 2 — longitudinal wave, reflection, Snell's law, TIR — QP screenshot

20 Which diagram shows reflection by a plane mirror?



21 Light is incident on a plastic block of refractive index 1.5.

The angle of incidence is 50° .

What is the angle of refraction?

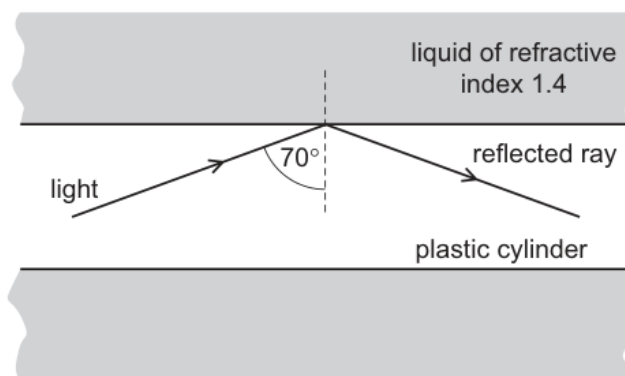
A 31°

B 33°

C 40°

D 75°

22 A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70° . The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

**F: O/N 2017 Paper 1 Variant 2 — longitudinal wave, reflection, Snell's law, TIR
— respective MS screenshot**

Question	Answer	Marks
1	D	1
2	A	1
3	A	1
4	C	1
5	C	1
6	C	1
7	D	1
8	A	1
9	D	1
10	C	1
11	B	1
12	A	1
13	B	1
14	B	1
15	C	1
16	B	1
17	A	1
18	D	1
19	A	1
20	A	1
21	A	1
22	C	1
23	B	1
24	C	1
25	A	1
26	B	1
27	B	1
28	D	1

G: O/N 2012 Paper 1 Variant 2 — ripple-tank reflection, plane mirror image, refraction/TIR — QP screenshot

- 20 A strip is made from two metals joined together. The diagrams show the strip at room temperature and after it has been cooled.



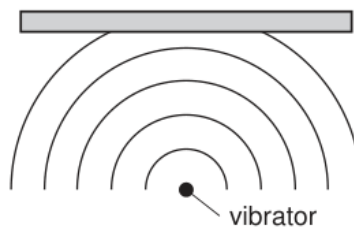
at room temperature



below room temperature

The change in shape occurs because

- A brass contracts more than invar.
 - B brass expands when it cools down.
 - C invar and brass contract by equal amounts.
 - D invar contracts more than brass.
- 21 In a ripple tank, a vibrator produces circular wavefronts which hit a flat surface.



The reflected wavefronts are also parts of circles. Where is the centre of these circles?

A ●

B ●

C ●

D ● vibrator

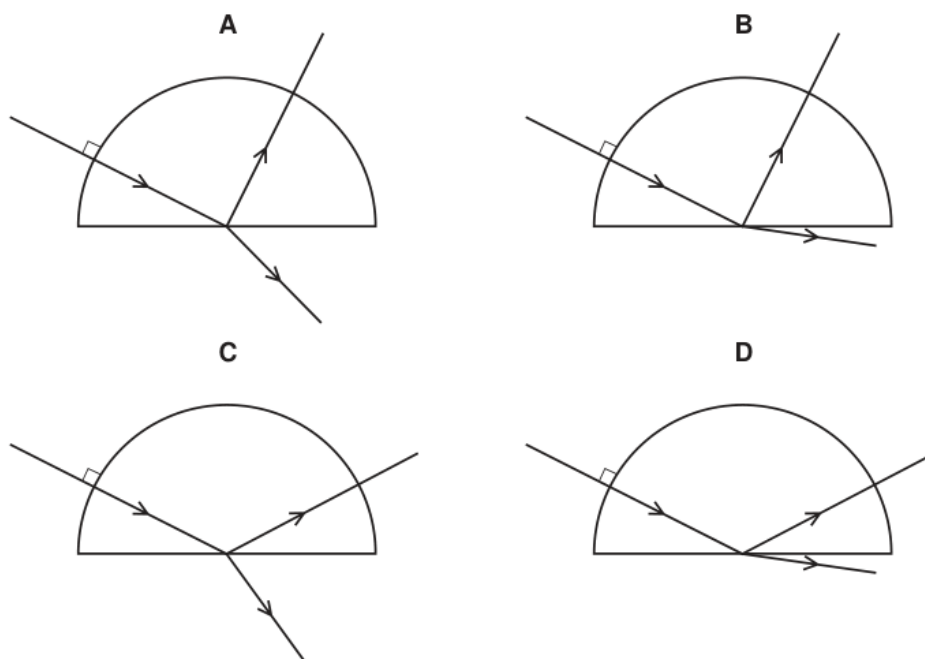
- 22 An object is placed in front of a plane mirror. The image produced is

- A real and smaller than the object.
- B real and the same size as the object.
- C virtual and smaller than the object.
- D virtual and the same size as the object.

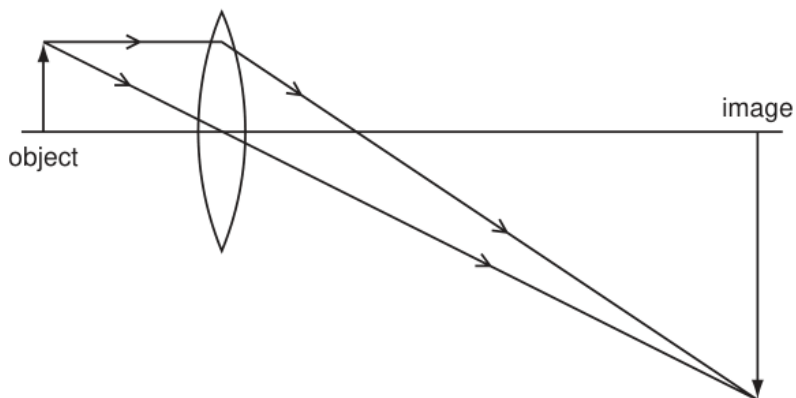
G: O/N 2012 Paper 1 Variant 2 — ripple-tank reflection, plane mirror image, refraction/TIR — QP screenshot

- 23** A ray of red light enters a semi-circular glass block normal to the curved surface.

Which diagram shows the partial reflection and refraction of the ray?



- 24** A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification of the object?

- A** 0.33 **B** 3.0 **C** 4.0 **D** 6.0
- 25** Which statement about red light and blue light is correct?
- A** Red light has a higher frequency than blue light.
B Red light has a longer wavelength than blue light.
C Red light has the same speed in glass as blue light.
D Red light is refracted by a glass prism more than blue light.

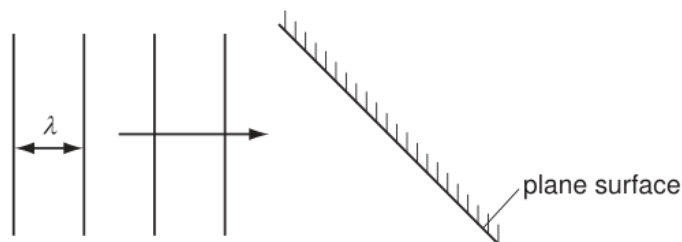
G: O/N 2012 Paper 1 Variant 2 — ripple-tank reflection, plane mirror image, refraction/TIR — respective MS screenshot

Page 2	Mark Scheme	Syllabus Paper
	GCE O LEVEL – October/November 2012	5054

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	21	A
2	C	22	D
3	A	23	D
4	A	24	B
5	D	25	B
6	C	26	C
7	D	27	A
8	C	28	C
9	B	29	B
10	D	30	B
11	B	31	A
12	D	32	C
13	D	33	D
14	A	34	D
15	C	35	C
16	C	36	C
17	A	37	B
18	B	38	A
19	B	39	C
20	A	40	C

H: O/N 2010 Paper 1 Variant 1 — ripple tank reflection and refraction by glass — QP screenshot

- 20 What is caused by the thermal expansion of a substance when heated?
- A a decrease in the resistance of a tungsten-filament lamp when switched on
 - B a rise in the pressure of the gas trapped in a gas cylinder placed in hot water
 - C the blowing of the fuse in a circuit when the current becomes too large
 - D the upward movement of the air above a Bunsen burner when it is lit
- 21 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	faster	shorter
B	slower	shorter
C	the same	longer
D	the same	the same

- 22 A ray of light strikes the surface of a glass block at an angle of incidence of 45° .

The refractive index of the glass is 1.5.

What is the angle of refraction inside the block?

- A 28°
 - B 30°
 - C 45°
 - D 67°
- 23 An object is viewed through a concave (diverging) lens.

What is the correct description of the image formed?

- A real, inverted, magnified
- B real, upright, diminished
- C virtual, inverted, magnified
- D virtual, upright, diminished

**H: O/N 2010 Paper 1 Variant 1 — ripple tank reflection and refraction by glass
— respective MS screenshot**

Page 2	Mark Scheme: Teachers' version	Syllabus Paper
	GCE O LEVEL – October/November 2010	5054

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	C	21	D
2	D	22	A
3	B	23	D
4	C	24	B
5	C	25	C
6	D	26	C
7	A	27	B
8	B	28	B
9	B	29	A
10	A	30	B
11	D	31	B
12	A	32	D
13	A	33	A
14	D	34	B
15	B	35	D
16	C	36	A
17	A	37	A
18	B	38	A
19	D	39	A
20	D	40	C